SOHANA DAS

Computer Science & Engineering | AI / Deep Learning, Full-Stack & Cyber Systems

Nagpur, Maharashtra, India • iamsohanadas1408@gmail.com • +91 8777741907 • linkedin.com/in/sohana-das-943a44311



PROFESSIONAL SUMMARY

High-performing **Computer Science & Engineering undergraduate (CGPA: 8.48)** and published patent co-inventor with hands-on internship experience at **Maincrafts Technology** and certified simulations from **Tata & Deloitte**. Proficient in **Machine Learning, NLP (BERT, GPT, Transformers, LSTM), Full-Stack Development, and Cybersecurity**. Proven track record in developing AI-powered chatbots, end-to-end NLP pipelines, and scalable database architectures. Actively seeking AI/ML, Data Science, and Software Engineering internships and full-time roles.

INTERNSHIP EXPERIENCE

Maincrafts Technology ↗ 2025

Software Development & AI Intern **Certificate ↗**

- Developed and integrated backend logic and AI-powered conversational chatbot modules for data-driven applications.
- Engineered RESTful API endpoints and optimized database schemas in Python and SQL, improving query efficiency and data handling.
- Collaborated on software testing, code modularization, and web UI integration using modern web standards.

PATENTS & PUBLICATIONS

Patent Published: ML & DL-Based Fake News Detection System ↗ Indian Patent Office (2026)

Application No.: 202521126334 | **Role:** Co-Inventor & Lead Pipeline Architect

- Architected an end-to-end automated NLP detection pipeline combining **contextual text embeddings, semantic analysis, and deep neural classifiers** to identify misinformation across digital media.
- Designed preprocessing, vectorization, and multi-class classification workflows published in the **Indian Patent Office Journal (2026)**.

TECHNICAL & RESEARCH PROJECTS

Fake Review Detection using Deep Learning Models

2025 – Present

Tech Stack: Python, PyTorch/TensorFlow, BERT & RoBERTa, Bi-LSTM, Hugging Face, Flask API, Scikit-Learn

- Engineered an advanced deep learning framework to detect fraudulent and bot-generated reviews across **50,000+ e-commerce records**.
- Implemented fine-tuned **Transformer models (BERT, RoBERTa)** paired with **Bidirectional LSTMs and attention mechanisms**, boosting classification accuracy to **94.8% (0.93 F1-score)** over baseline classifiers.
- Built an end-to-end REST inference API with **Flask** for real-time review sentiment scoring and spam probability verification.

Contextual Embedding for Next Word Prediction

Jan 2025 – Jun 2025

Tech Stack: Python, PyTorch, GPT-2, BERT, LSTM, GRU, Penn Treebank Dataset, NLP

- Designed, trained, and benchmarked recurrent architectures (**LSTM, GRU**) alongside transformer models (**BERT, GPT-2**) for sequence prediction.
- Optimized contextual embedding layers on the benchmark **Penn Treebank dataset**, achieving a **28% reduction in perplexity** and accelerating inference.

Fake News Detection using Machine Learning & Deep Learning

Jul 2025 – Dec 2025

Tech Stack: Python, SVM, LSTM, BERT, TF-IDF, NLTK, Scikit-Learn

- Developed an automated news verification engine utilizing SVM, LSTM, and pre-trained BERT on a curated corpus of **44,000+ labeled news articles**.
- Formulated custom NLP feature extraction pipelines (n-grams, token embeddings) and evaluated models across **Precision, Recall, and F1-score metrics**.

TECHNICAL SKILLS & COMPETENCIES

Languages:	**Python**, **Java**, **C**, **SQL** (PL/SQL), JavaScript (ES6+), HTML5, CSS3
AI / ML & NLP:	**PyTorch**, **TensorFlow**, **BERT**, **GPT-2**, **Transformers**, **LSTM**, GRU, SVM, Scikit-Learn, Hugging Face
Databases & Cloud:	**PostgreSQL**, **MySQL**, **Oracle Database (PL/SQL)**, SQL Plus, Relational Modeling
Web & Frameworks:	**React.js**, **Flask**, RESTful APIs, Git, GitHub, Jupyter Notebook, VS Code, Linux
Cyber & Core CS:	**Identity & Access Management (IAM)**, Threat Assessment, **Data Structures & Algorithms**, OOP, OS, DBMS

CERTIFICATIONS & INDUSTRY CREDENTIALS

Software & AI Internship Certificate ↗
Maincrafts Technology
AI chatbot development, backend logic, and scalable data-driven application workflows.
Completed 2025 — Certificate ↗

Cybersecurity Analyst Job Simulation ↗
Tata (via Forage)
IAM fundamentals, enterprise strategy assessment, and custom IAM solutions.
June 2026 — Certificate ↗

Cyber Job Simulation ↗
Deloitte (via Forage)
Enterprise cybersecurity threat modeling, security governance, and risk mitigation.
June 2026 — Certificate ↗

Introduction to Python Programming ↗
Red Hat Academy
Python data structures, OOP paradigms, modular architecture, and algorithms.
February 2025 — Certificate ↗

IBM SkillsBuild Technical Foundations ↗
IBM SkillsBuild
February 2025 • Certificate ↗

EDUCATION

B.Tech in Computer Science & Engineering — CGPA: 8.48 / 10
Symbiosis Institute of Technology, Nagpur
July 2024 – June 2028

Class XII (ISC) & Class X (ICSE) — XII: 84.5% | X: 93.2%
St. Claret School, Kolkata
2022 – 2024 | Passed 2022